

Audio course on steroids during pregnancy. Episode three.

Antenatal corticosteroids for fetal lung maturation.

Companion reading handout for simultaneous listening.

This episode is about the clearest fetal-benefit use of steroids in pregnancy: antenatal corticosteroids before likely preterm birth.

The clinical tension is simple. A preterm fetus may be born before the lungs and other organs are ready for extrauterine life. We have a short window where maternal treatment can improve neonatal outcomes. But the treatment only makes sense if birth is likely soon enough, and if delaying delivery to complete the course is safe.

So this is not just a drug order. It is a timing decision.

Betamethasone and dexamethasone are chosen because they cross the placenta. In this setting, fetal exposure is intentional. These drugs accelerate fetal maturation, especially lung maturation. They improve surfactant-related pathways and help prepare the newborn for air breathing.

The benefits of a correctly timed course in the early preterm setting are among the most important in obstetrics: lower respiratory distress syndrome, lower intraventricular hemorrhage, lower necrotizing enterocolitis, and lower neonatal mortality.

Now the regimens.

Betamethasone is twelve milligrams intramuscularly, repeated once after twenty four hours.

Dexamethasone is six milligrams intramuscularly every twelve hours for four doses.

Do not teach these as isolated numbers. Teach what they represent. They are short, deliberate fetal exposures designed for a fetus likely to be born preterm.

The classic window is twenty four weeks to thirty three weeks and six days when delivery within seven days is likely. ACOG and S M F M also allow consideration at twenty two weeks to twenty two weeks and six days if neonatal resuscitation is planned after appropriate counseling. At twenty three weeks, practice also depends heavily on periviability counseling, neonatal intent, and local outcomes.

The seven-day window matters.

Antenatal steroids are not a background supplement for every high-risk pregnancy. The benefit is strongest when birth occurs soon after the course. If we give steroids too broadly to patients who remain pregnant for weeks or deliver at term, we expose many fetuses without the intended neonatal benefit.

That is the first major resident mistake: giving steroids because the pregnancy feels high risk, without asking whether birth is actually likely soon.

The second mistake is delaying necessary delivery just to complete steroids.

If the mother has a severe condition requiring delivery, or the fetus is compromised, steroid completion cannot become the organizing goal. The fetus benefits from maturity only if the mother and fetus can safely remain pregnant long enough. Steroids improve the preterm transition. They do not make an unsafe pregnancy safe.

Now consider partial treatment.

The Israeli preterm labor position paper notes that partial treatment can still be effective. If birth occurs before the full course is complete, the doses already given may still help. But that is not the same as shortening the dosing interval because we are anxious. There is no reason to compress the standard interval just because delivery is expected within twenty four hours.

This distinction matters in the delivery room.

The resident should not say, we cannot give steroids because there is no time for a full course. If there is time to give a dose and delivery is not immediate, a partial course may still be worth giving.

But the resident should also not say, let us speed up the doses. The regimen is the regimen.

Now let us connect antenatal steroids to tocolysis.

Tocolysis does not cure preterm birth. It buys time. Often the time being purchased is forty eight hours for steroids, magnesium neuroprotection if indicated, G B S prophylaxis, and maternal transfer to a center with appropriate neonatal care.

That is a useful purpose. But it is not always safe. Infection, significant bleeding, severe preeclampsia, fetal compromise, or maternal instability can make tocolysis inappropriate. A resident should never think of tocolysis as an automatic steroid-completion tool.

Now think about fetal monitoring.

After betamethasone or dexamethasone, fetal behavior can transiently change. Fetal breathing movements and heart rate reactivity may decrease for a short period. This can create false concern on NST or BPP interpretation.

The teaching point is not to ignore fetal testing. The point is to interpret it in context. A fetus after antenatal steroids may temporarily look less active or less reactive. Escalation should be based on the whole clinical picture, not a reflexive response to a steroid-related transient change.

Now let us test the model.

A patient is thirty weeks with regular contractions, cervical change, no infection, reassuring fetal status, and no contraindication to short delay. This is a classic steroid moment. The fetus is the target, preterm birth is plausible in the next week, and time appears safe.

Another patient is thirty four weeks with severe preeclampsia, persistent neurologic symptoms, and severe-range BP requiring urgent management. Steroids may be relevant if delivery is preterm, but steroid completion is not the main objective. Stabilize the mother. Do not turn fetal maturation into a reason to delay medically indicated delivery.

Another patient is twenty eight weeks with vague abdominal discomfort but no contractions, no cervical change, and no concrete reason to expect delivery soon. The pregnancy may feel high risk, but the seven-day logic is missing. Steroids should not be used as a vague insurance policy.

The resident reflex is this: antenatal corticosteroids are timed fetal therapy for likely preterm birth, not a background medication.

Use betamethasone or dexamethasone because the fetus is the target.

Use them when birth is likely soon enough that the fetus can benefit.

Give the standard regimen without inventing shortcuts.

Do not delay unsafe delivery for steroid completion.

And when interpreting fetal testing after steroids, remember that the drug can transiently change fetal behavior.

That is how an extremely powerful therapy remains precise.